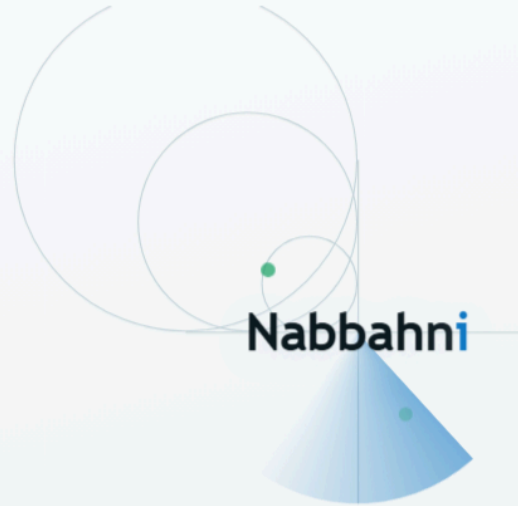




HELLO. I AM NABBAHNI. THE WEATHER FORECAST FOR MEDICINES. █

LA MÉTÉO DES MÉDICAMENTS • DON'T DETECT SHORTAGES. PREVENT THEM. • AUTOMATE OR DIE — HEALTHCARE & PHARMA



THE PROBLEM: PATIENTS WALK FROM PHARMACY TO 

PATIENTS

They search, pharmacy after pharmacy

A sick person visits 3, 4, 5 pharmacies to find one medicine. Sometimes they give up.

PHARMACIES

They find out too late

The shelf is already empty when the shortage is discovered. Too late to order —

CENTRAL PHARMACY

It reacts after the crisis

Reports arrive

OUR QUESTION



MY SOLUTION: PREDICT. EXPLAIN. ■

MODULE 1

Data Hub

All the data in one place:
pharmacy stocks, **daily sales**, PCT
central stock, supplier delays,
weather and epidemics.

MODULES 2-3

Demand AI + Shortage AI

I predict consumption — insulin:
2,000 → 2,700 units in 3 weeks
— and shortage

MODULE 4

Explainable AI

I give the causes: consumption

MODULE 5

POURQUOI ?

Euthyrox : risque de rupture 87 % — consom

تعداد



I AM CAREFUL. I NEVER LEARN THE WRONG > ■

CLEVER

I never learn the wrong lesson

When a medicine is out of stock, sales look low. I correct this — so I **never believe demand fell** during a shortage.

HONEST

I give a range, not one number

Best case, expected, worst case. You always see **how sure I am** —

FAST

My reasons are ready in advance

SAFE



MY BRAIN: REAL DATA. REAL AI. ■

EXTERNAL DATA • APIS

Connected to world reality

Global purchase KPIs, world manufacturers, **pandemics, wars, inflation**, raw-material shortages — live via APIs to world news. Real signals, **never generated**.

INTERNAL DATA

Every pharmacy, every sale

Real-time stocks, daily sales, orders and **supplier delays** from pharmacies and the Central Pharmacy.

ML • TRAINING

XGBoost • LightGBM • Prophet

Demand forecasts per

LLM • REASONING



MY BUSINESS MODEL: THE STATE SAVES. ■

TAM $\approx$ 30 M TND/an – Maghreb & Afrique francophone

SAM $\approx$ 2,3 M TND/an – Tunisie

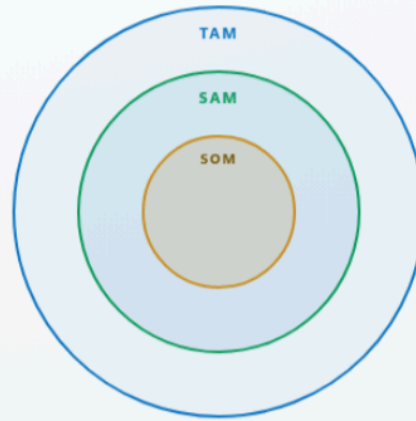
SOM an 1 $\approx$ 250 k TND –

B2G • CORE REVENUE

B2B • PHARMACIES

DATA • API

SCALE

**TAM • MAGHREB + AFRICA****~30 BN TND**

Pharmaceutical spend across francophone Africa. Every one of these markets lives the same shortage and expiry problem.

SAM • TUNISIA**~8.5 BN TND**

Tunisian pharmaceutical market — **2.74 BN USD** in 2025, heading to **6.4 BN USD** by 2032. Served by **2,000+ pharmacies**, 60+ wholesalers, 40 plants.

SOM • THE LOSS WE ATTACK**~250–420 M TND**

At a conservative **3–5% waste rate**, that is what expires unsold every year. My own projection on the pilot dataset is **8.5%**. We do not need to invent a market — we take a fee on money already being burned.

Sources: Maximize Market Research 2025 • CNOPT • DPM • L'Économiste Maghrébin.
Waste range is an industry-standard estimate, stated as such.



THE MARKET IS NOT A GUESS. I SIZED TH1

B2G • ANCHOR**National license****600 k**

TND / year • PCT + Ministry

Under **0.25%** of the annual waste it targets. One avoided emergency import pays for it.

B2B • VOLUME**Pharmacy Pro****40**

TND / month • per pharmacy

Alerts stay free. Pro adds order sizing and substitution. **300 of 2,000** offices at year 3 = **144 k TND**.

DATA • MARGIN**Supplier API****25 k**

TND / year • per subscriber

Anonymised demand forecasts for importers and manufacturers. **8 subscribers = 200 k TND**.

SCALE**New countries****×N**

same platform, new feeds

Marginal cost of a new country is a data connector, not a rebuild.

**FOUR REVENUE STREAMS. ONE PAYS FOR THE OTHERS.****0.6 M**

YEAR 1 • PILOT

0.9 M

YEAR 2 • NATIONAL

1.4 M

YEAR 3 • TND ARR



MY IMPACT: SHORTAGES PREDICTED 15 TO 30 DA/\

EARLY

15–30 days of warning

Enough time to import or redistribute **before shelves are empty** — instead of reacting after.

WORKLOAD

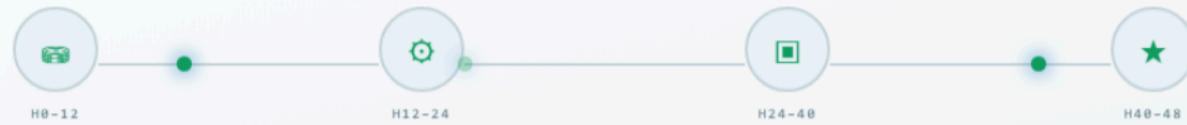
50% less pharmacist workload

No more calling around to find medicines. Substitutes appear in one click.

CITIZENS

1 search instead of 5 pharmacies

MONEY & WASTE



FEASIBILITY: I WAS BUILT IN 48 HOURS WITH F~+

HOURS 0-12

Build the data

We generate a realistic Tunisia:
24 regions, 68 medicines, 2
years of daily history.

HOURS 12-24

Train the AI

We train and test the prediction
models. Normal laptops — no
GPU needed.

HOURS 24-40

Build the app

API, national map,

HOURS 40-48

	EXCEL + ERP	GENERIC BI	MED-FINDER APPS	NABBAHNI
Predicts shortages	×	×	×	✓
Explains its alerts	×	×	×	✓
French + Arabic	×	×	~	✓
State + pharmacy + citizen	×	~	~	✓



OTHERS SHOW THE PAST. I PREDICT THE F>

EXCEL REPORTS

They describe last month

Numbers arrive weeks late. I warn about **next month** — with reasons.

BI DASHBOARDS

Beautiful charts of the past

No prediction, no explanation, no Arabic. I am the **layer above** them.

MED-FINDER APPS

They help one patient today

|

● Running today

- **FastAPI backend** — 40+ endpoints, RBAC on 6 roles, audit trail
- **Next.js frontend** — 29 pages, 28 .tsx components, fr/ar with RTL
- **Trained models** — 4 horizons, WAPE **6.0%**, versioned in the repo
- **SHAP explanations** — 1,700 predictions, all explained
- **Expiry + allocation** — 9,147 lots, FEFO, 627 k TND recoverable
- **28 automated tests** · one-command local launch

● Designed, not yet wired

- **Kubernetes + Keycloak + RabbitMQ** — manifests and realm written, disabled in the demo
- **PostgreSQL + PostGIS** — schema is dialect-portable; the demo runs on SQLite so it starts anywhere in 8 minutes
- **External API feeds** — World Bank and openFDA connectivity verified, not yet ingested
- **LLM layer** — interface defined, no key wired



WHAT IS **ACTUALLY BUILT** — AND WHAT IS ONLY DES

Training data is **synthetic** — metrics are pipeline proof, not field performance

AUC **0.999** is an artefact of the generator, and we say so



DEMO TIME. WATCH ME SAVE **INSULIN** —

STEP 1**Data goes in**

Stocks, sales and supplier data are **injected** into the platform.

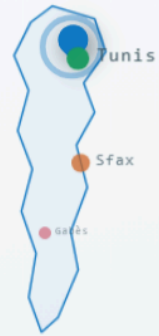
STEP 2**I detect**

Risk of insulin shortage **in 35 days**.

STEP 3**I explain**

Consumption **+18%** · supplier late

STEP 4**STEP 5****STEP 6**



GIVE ME ONE REGION AND ONE FLU SEASON. █

15-30 days of early
warning

7 of 10 shortages caught in
advance

20 actions validated by the Central
Pharmacy